$^{37}Cl(^{11}B,^{13}N)$ 1988Or01

1988Or01: An 81-MeV $^{11}B^{5+}$ beam was produced from the ANU 14UD Pelletron accelerator. Targets were enriched $BaCl_2$. Reaction products were momentum-analyzed with an Enge split-pole spectrometer (FWHM=200 keV at 5.75-10.25°) and detected with a multi-element gas-filled detector at the focal plane. Measured $\sigma(E(^{13}N))$. Deduced levels. Comparisons with shell-model and DWBA calculations. Proposed a decay scheme of ^{35}Si based on the γ transitions observed by 1986Du07.

^{35}P Levels

E(level)†	Comments
0	$d\sigma/d\Omega$=120 μb/sr *30*.
2389 *4*	$d\sigma/d\Omega$=200 μb/sr *50*.
3860 *10*	$d\sigma/d\Omega$=35 μb/sr *9*.
4250 *20*	
4640 *20*	$d\sigma/d\Omega$=30 μb/sr *8*.
5010 *20*	
5220 *40*	$d\sigma/d\Omega\approx$10 μb/sr *3*.
5840 *50*	
7590 *20*	$d\sigma/d\Omega$=45 μb/sr *11*.
8390 *40*	

† From 1988Or01.